
BudgetFlow: Value-Aware Budget Governance for Agent Tasks

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Abstract

Multi-step AI agent tasks consume variable amounts of model budget, and the tasks in a batch often have unequal value. Existing model-routing work primarily decides which model should answer one request or how one task should spend its own budget. We study *Batch-Level Task Budget Allocation*: a task batch shares one hard budget, and the system must allocate model opportunities across tasks before the budget is exhausted.

We introduce BudgetFlow, a value-aware budget governance framework for task batches. Its interface combines task-level signals—Task Value, Estimated Token Cost, Model Fit, and a Verifier—with batch-level Shared Budget state. BudgetFlow uses the first three task signals and the Shared Budget to allocate budget and model tiers, while the Verifier settles outcomes. We evaluate BudgetFlow on SWE-bench-style verified bug-fixing tasks under fixed task sets and shared hard budgets. The paper reports three primary metrics: Resolved Rate, Total Resolved Value (TRV), and TRV per Dollar.

Across one 4-policy run and three 5-policy runs over 30-task batches, BudgetFlow provides both a policy and a frontier-reporting interface. The framework produces the latest audited run’s best Resolved Rate, TRV, and TRV per Dollar; it also exposes a value-efficiency gain in an earlier run where raw resolved count favors the strong-model-only baseline. Sensitivity analyses over Task Value, budget cap, and KV-cache cost discount show how the cost-value frontier moves as value, budget pressure, and serving costs change. These results support BudgetFlow as a concrete formulation of value-aware task-batch budgeting, with SWE-bench as a verified coding instantiation.

1. Introduction

AI agents increasingly execute multi-step tasks: they inspect context, call tools, edit artifacts, run checks, and decide whether another model call is worth the cost. These tasks consume budget unevenly. One task may finish after a short reasoning loop; another may require repeated search, repair, and verification. In a batch, this variation means that a fixed model budget can be exhausted before every task receives

the same opportunity to succeed.

A central issue is that tasks rarely have equal value. A support ticket with an imminent service-level agreement, a security patch, a revenue-critical spreadsheet fix, and a routine rewrite should receive different budget treatment when they are submitted in the same batch. Once model calls become a scarce shared resource, the system needs an allocation policy in addition to a model router.

This paper studies *Batch-Level Task Budget Allocation*. The general setting has many tasks, one shared hard budget, model tiers with different costs and capabilities, and an outcome rule that decides whether a task is resolved. The objective is to allocate scarce model opportunities so the batch produces more verified task value under the same resource constraint.

The operational decision is simple: which task should receive the next scarce model opportunity, under the same Shared Budget?

BudgetFlow is a budget governance framework for this setting. It uses four task-level signals and one batch-level budget state:

- **Task Value:** the pre-registered value gained if the task is resolved;
- **Estimated Token Cost:** the pre-run estimate of token cost used for cap planning;
- **Model Fit:** evidence about how well each model tier fits the task;
- **Verifier:** the outcome rule that maps a completed attempt to resolved or unresolved;
- **Shared Budget:** the hard batch-level resource constraint shared by all tasks.

The first three task-level signals and the Shared Budget drive allocation decisions. The Verifier settles outcomes and defines the objective. This separation is the core interface. The policy can change how it routes, caps, assigns model opportunities, or stops tasks, while the outcome contract

remains explicit. Task-source labels such as coding, marketing, support, or document production can be recorded as context, but they are not themselves the decision rule for strong-model access.

We instantiate this formulation on SWE-bench-style verified coding tasks. SWE-bench is useful here because it gives a strict task-batch testbed: each task is a real bug-fixing issue, each attempt has measurable model spend, and each outcome is settled by a test-based harness. The benchmark is the evidence domain, while the allocation interface is the research object.

The paper uses three primary metrics. **Resolved Rate** anchors the evaluation to standard verified success. **Total Resolved Value** (TRV) sums pre-registered Task Value over resolved tasks. **TRV per Dollar** measures value efficiency under the shared hard budget.

The central empirical result is the BudgetFlow frontier view. BudgetFlow wins the latest audited 5-policy, 30-task run on Resolved Rate, TRV, and TRV per Dollar. It also wins a 4-policy run on value and value efficiency while resolving one fewer task than the strong-model-only baseline. Strong-model-led runs are informative within this same framework: they mark operating conditions where broad use of the strong model is already a strong frontier point. Together, these outcomes show the frontier created by Shared Budget pressure, task value, and model-tier fit.

This paper makes three contributions:

1. We formulate Batch-Level Task Budget Allocation with pre-registered Task Value, Estimated Token Cost, Model Fit, a shared hard budget, and a Verifier-settled objective.
2. We present a generalizable BudgetFlow interface that allocates model opportunities using Task Value, Estimated Token Cost, Model Fit, Shared Budget, and a Verifier-settled outcome contract.
3. We evaluate BudgetFlow against cheap-model-only, strong-model-only, learned task-router, and budget-only baselines, reporting Resolved Rate, TRV, TRV per Dollar, and frontier modes driven by Value Divergence, Strong Coverage, and Model Fit.

2. Related Work

Per-query routing and cascading. RouteLLM learns per-query routers from preference data (Ong et al., 2025). Cascade Routing studies model routing and cascading for individual queries (Dekoninck et al., 2025). RouteNLP targets enterprise query routing under quality constraints (Guo et al., 2026). UCCI calibrates uncertainty for cost-optimal cascade routing (Kotte, 2026). These works make model selection

a first-class cost-quality problem. The unit is typically one prompt or one request. BudgetFlow operates at the task-batch layer: many multi-step tasks share one hard budget, and the policy allocates scarce model opportunities across tasks with different values.

Within-task budgeted agents. INTENT studies budget-constrained tool use inside a single agent task (Liu et al., 2026). This is a complementary layer to Batch-Level Task Budget Allocation. A within-task policy decides whether one task should spend another tool or model call. BudgetFlow decides how one budget should be shared across many tasks before the batch budget is exhausted.

Coding-agent evaluation and cost-aware runtimes. SWE-bench provides the verified coding-task substrate used by many agent evaluations (Jimenez et al., 2024). OpenSquilla is a token-efficient agent runtime with model routing, diagnostics, replay, and cost-per-success framing (OpenSquilla contributors, 2026). Claw-SWE-Bench studies harnesses, adapters, prompts, patch extraction, leakage cleanup, and cost reporting for SWE-bench-style coding-agent evaluation (Zheng et al., 2026). These systems sharpen the evaluation requirements for agent tasks: cost, harness behavior, and outcome verification must be explicit.

Serving substrates. vLLM improves LLM serving through PagedAttention and KV-cache management (Kwon et al., 2023); SGLang accelerates structured language-model programs with runtime optimizations such as KV-cache reuse (Zheng et al., 2024); Dynamo exposes agentic inference concerns including cache locality and scheduling (NVIDIA, 2026). These systems provide execution substrates. BudgetFlow provides task-level value and budget signals that can inform priority, continuation, model opportunity, and cache-sticky scheduling in future serving-aware systems.

3. Problem Formulation

Let a task batch contain tasks $\mathcal{T} = \{1, \dots, n\}$ and let B denote the Shared Budget. Each task i has a pre-registered value $v_i > 0$, an estimated token cost $d_i > 0$, and model-fit evidence $m_{i,k}$ for each model tier k . A policy π chooses a sequence of actions that spend budget on model calls and task attempts. Let $c_i(\pi)$ be the spend on task i , and let

$$\sum_{i=1}^n c_i(\pi) \leq B$$

be the shared hard-budget constraint.

The budget is shared at the batch level. A policy can spend more on one task only by leaving less budget for later tasks

in the same batch. This coupling is what separates the problem from independent per-task routing: a locally reasonable strong-model call can be globally poor if it consumes budget needed by a higher-value task.

Each task has a Verifier V_i that maps the final task evidence to a binary outcome:

$$r_i(\pi) = V_i(\pi, i) \in \{0, 1\}.$$

The primary value objective is Total Resolved Value:

$$\text{TRV}(\pi) = \sum_{i=1}^n v_i r_i(\pi).$$

We report three headline metrics:

$$\text{ResolvedRate}(\pi) = \frac{1}{n} \sum_{i=1}^n r_i(\pi),$$

$$\text{TRVperDollar}(\pi) = \frac{\text{TRV}(\pi)}{\sum_i c_i(\pi)}.$$

3.1. Core Terms and Metric Reading

Table 1 gives the compact reading guide used throughout the paper. The metrics are intentionally simple: success is verifier-settled, value is pre-registered, and efficiency is value divided by spend. This makes the frontier interpretable through standard verified outcomes and a pre-registered value ledger.

Term	Reading
Batch-Level Task Budget Allocation	A task batch shares one hard budget; a 30-task run spends at most \$9.95 total.
Task Value	Pre-registered value gained when the task passes the Verifier.
Estimated Token Cost	Pre-run token-cost estimate used for cap planning.
Model Fit	Evidence about which model tier is likely to help this task.
Verifier	Domain outcome rule; SWE-bench uses a test-based harness.
Shared Budget	Hard batch-level resource constraint shared by all tasks.
Resolved Rate	Verifier-passed tasks divided by all tasks; 16/30 means 53.3%.
Total Resolved Value (TRV)	Sum of Task Value over verifier-passed tasks.
TRV per Dollar	TRV divided by total spend; TRV 18.0 at \$9.95 gives 1.81 TRV/\$.

Table 1. Core terms and metrics. The examples show how to read the reported numbers.

The Verifier is an outcome settlement rule. The policy uses Task Value, Estimated Token Cost, Model Fit, and Shared Budget state for allocation; the Verifier settles the final

outcome after execution. In our SWE-bench instantiation, V_i is the test-based harness. This gives a strict external success signal while preserving the task-batch allocation structure.

The interfaces have different portability requirements. Task Value must be fixed before execution. Estimated Token Cost is a pre-run planning estimate; final spend is measured after execution. Model Fit can come from catalog priors, historical outcomes, or calibration runs. The Verifier must be a trusted outcome rule for the domain. The Shared Budget is set at the batch level. Their concrete calibration changes by domain, but the allocation problem consumes the same interface.

4. BudgetFlow

BudgetFlow maintains a Shared Budget ledger across the task batch. For each task, it estimates whether the next model opportunity should use the cheap model, the strong model, or defer additional spend. The ledger is hard: Shared Budget gates each planned action.

The policy uses three allocation inputs:

- v_i , the Task Value;
- d_i , the Estimated Token Cost;
- $m_{i,k}$, the Model Fit for tier k .

These inputs enter distinct decisions. Task Value prioritizes tasks that create more value if verified. Estimated Token Cost becomes cap planning: the policy estimates how much token cost a task may need before it can produce a useful artifact. Model Fit estimates whether the strong model is likely to improve the outcome relative to the cheap model. Shared Budget controls affordability and stop/defer decisions.

Let h denote budget pressure, with higher h indicating tighter Shared Budget state. Let k_c be the cheap model and k_s be the strong model. A simplified task-start score for giving task i a strong-model opportunity is

$$S_i = \frac{v_i \cdot \max(m_{i,k_s} - m_{i,k_c}, 0)}{\max(d_i, \epsilon)} \cdot q(h),$$

where d_i is the task’s Estimated Token Cost and $q(h)$ reduces strong-model access as Shared Budget tightens. The implementation uses this score with affordability checks, learned task-router priors, and hard-budget gates. The formula captures the core allocation principle: strong-model budget is most valuable when a task has high value, sufficient expected model-fit gain, manageable Estimated Token Cost, and enough Shared Budget.

Interface	SWE-bench instantiation	Other task-batch instantiations
Task Value	pre-registered criticality value	business priority, user tier, task owner value
Estimated Token Cost	pre-run token-cost estimate from task metadata	expected context size, tool steps, artifact size
Model Fit	tier evidence from prior runs and catalog priors	domain-specific success prior by model tier
Verifier	test-based harness	human review, rule check, delayed business outcome
Shared Budget	batch-level cap in dollars	team budget, project budget, processing window budget

Table 2. Domain interfaces for Batch-Level Task Budget Allocation. The paper evaluates the SWE-bench instantiation.

Algorithm 1 BudgetFlow batch policy sketch

- 1: **Initialize** Shared Budget state $R \leftarrow B$.
- 2: **for** task i in the batch **do**
- 3: **Read** $(v_i, d_i, m_{i,k_c}, m_{i,k_s})$ and current R .
- 4: **Set** an initial task cap from Task Value, Estimated Token Cost, and R .
- 5: **Choose** cheap or strong model from Model Fit, value upside, and affordability.
- 6: **Execute** the task while updating spend and Shared Budget.
- 7: **Continue, stop, or defer** using value upside and budget pressure.
- 8: **Settle** the final artifact with V_i and record r_i .
- 9: **end for**
- 10: **Report** Resolved Rate, TRV, TRV per Dollar, and total spend.

BudgetFlow separates allocation from outcome settlement. It uses Task Value, Estimated Token Cost, Model Fit, and Shared Budget state to choose model opportunities. The Verifier evaluates the resulting patch or artifact after the task attempt. This design keeps the value objective auditable: the policy can explain why a task received budget, and the verifier independently determines whether that spend produced resolved value.

Allocation decisions. At task start, BudgetFlow assigns a route and an initial cap from value, Estimated Token Cost, Model Fit, and Shared Budget. During execution, it can assign a strong-model opportunity when the strong model has expected upside and Shared Budget allows it. It can stop or defer when additional spend has low expected value or would threaten the batch budget. These decisions are task-level in the current paper; finer stage or segment routing is left for future work.

Algorithm 1 is the implementation-facing view of the framework. The policy reads the three allocation interfaces before and during execution, while the Verifier is applied after the task attempt. This keeps routing, cap planning, and strong-model opportunity decisions auditable under the same value ledger used for evaluation.

5. Experimental Setup

Task set. We evaluate on fixed 30-task SWE-bench-style bug-fixing batches. Each task requires an agent to inspect a repository, edit code, and produce a patch. Outcomes are scored by the local harness and cross-checked with exported official predictions where available. Task values are pre-registered before execution.

Policies. Table 3 compares the five policies by the inputs they use. All policies run under the same shared hard budget and the same verifier. The key contrast is whether Task Value enters allocation of scarce model opportunities.

The 4-policy run uses the four policies available in that experimental line. All mainline policies use the same task order, budget protocol, model-tier catalog, and scoring path inside a run.

Metrics. The main tables report Resolved Rate, TRV, and TRV per Dollar, as defined in Table 1. Resolved Rate is written as both count and percentage. Spend is included as budget context because TRV per Dollar depends on it.

Sensitivity analyses. The main paid runs show observed frontier points. The sensitivity analyses ask how that frontier changes when the value ledger, the budget cap, or the serving cost model changes while the completed outcomes stay fixed. Task Value sensitivity re-scores fixed outcomes under alternate value profiles. Budget sensitivity replays the observed task order under different shared budget caps. KV Cache Cost-Discount sensitivity re-costs repeated input tokens under explicit cache-discount assumptions. Together, these diagnostics separate outcome quality from the accounting assumptions that determine cost-value efficiency.

6. Results

6.1. Main Paid Evidence

Table 4 summarizes four paid task-batch runs through the BudgetFlow frontier interface. Red entries mark BudgetFlow frontier signals; blue entries mark the strongest control when a control defines the observed frontier. The **latest**

Policy	Same budget	Task Value	Token-cost cap planning	Model choice	Same verifier
cheap-model-only	yes			fixed cheap model	yes
strong-model-only	yes			fixed strong model	yes
learned task-router	yes			value-blind learned choice	yes
budget-only	yes		yes	budget-pressure choice	yes
BudgetFlow	yes	yes	yes	value-aware choice	yes

Table 3. Policy inputs and shared outcome contract. Token-cost cap planning is the decision-facing use of Estimated Token Cost.

audited 5-policy run is the clearest positive policy result: BudgetFlow resolves 16/30 tasks, reaches TRV 18.00, and achieves 1.81 TRV per Dollar. The next-best learned task-router resolves 15/30 tasks, reaches TRV 17.00, and achieves 1.71 TRV per Dollar.

The runs show why BudgetFlow is useful beyond a single leaderboard row. First, value-aware allocation can protect high-value tasks and **improve TRV and TRV per Dollar** even when it resolves one fewer task than strong-model-only. Second, BudgetFlow exposes cases where **a strong model is already a strong frontier point** because it is both capable and cost-effective under the cap. Third, with the latest audited policy, BudgetFlow **improves all three primary metrics** against the five-policy table.

These outcomes should be read together. The 4-policy run is the clearest value-aware allocation case: raw resolved count alone favors the strong-model-only baseline, but the resolved tasks under BudgetFlow **carry more value and cost less**. The two 5-policy boundary runs show that strong controls are meaningful operating points inside the same frontier analysis. The latest audited 5-policy run shows the desired frontier movement: BudgetFlow **leads on standard resolution, value, and value per dollar** under the same cap.

6.2. Frontier Mode Analysis

Table 5 summarizes the mechanism exposed by each main operating regime. This analysis is not a separate metric family. It is the interpretation layer for the same Resolved Rate, TRV, and TRV per Dollar evidence in Table 4. The key point is that a shared-budget batch can be frontier-limited by different signals: value divergence, broad strong-model coverage, or accurate model-fit routing.

6.3. Full Latest 5-Policy Table

Table 6 gives the full latest audited run. **BudgetFlow leads all three primary metrics**. This is the clearest point-wise evidence for the current policy instantiation; the next section adds sensitivity curves to show how the same framework reports alternate operating conditions.

6.4. Task-Level Model Advantage

The latest run contains real model-tier heterogeneity. Among tasks where both cheap and strong model rows consumed model budget, there are **six cheap-model-favorable passes, five strong-model-favorable passes, and seven comparable failures under both**. This is the mechanism-level reason a batch-budget framework is useful: the task batch contains cheap-model opportunities, strong-model opportunities, and ceiling tasks in the same Shared Budget.

This task-level heterogeneity motivates selective strong-model use. When strong-model upside is concentrated on a subset of tasks, the allocation policy should preserve strong-model opportunities for those tasks while spending less on tasks where the cheap model is sufficient or both tiers are unlikely to pass.

7. Sensitivity Analysis

The sensitivity section has a different role from the main paid evidence. The main evidence reports what happened in completed runs. Sensitivity analysis asks whether the same completed outcomes still support the cost-value story when reasonable evaluation choices change. We therefore treat sensitivity as a frontier diagnostic: each table or curve explains how BudgetFlow behaves when value, budget pressure, or serving cost changes.

7.1. Task Value Sensitivity

Task Value sensitivity tests whether the latest positive result depends on one narrow value profile. Table 7 re-scores the latest audited run under alternate Task Value profiles. BudgetFlow remains ahead of the best control under equal value, criticality value, compressed criticality, and expanded criticality. A value permutation diagnostic over 64 permutations gives BudgetFlow a positive margin in all samples, with median margin +1.00 TRV. The conclusion is that the latest frontier movement is visible across several value ledgers, including the plain resolved-count view where all resolved tasks have equal value.

BudgetFlow: Value-Aware Budget Governance for Agent Tasks

Run	Leading policy	Resolved Rate	Spend	TRV	TRV per Dollar
4-policy value-aware allocation case	BudgetFlow strong-model-only	14/30 (46.7%)	\$7.98	18.50	2.32
		15/30 (50.0%)	\$9.46	18.00	1.90
5-policy strong-model boundary case	strong-model-only BudgetFlow	17/30 (56.7%)	\$9.95	20.00	2.01
		15/30 (50.0%)	\$9.95	17.50	1.76
5-policy learned-router stress case	strong-model-only BudgetFlow	18/30 (60.0%)	\$9.88	19.00	1.92
		17/30 (56.7%)	\$9.95	18.50	1.86
Latest audited 5-policy positive case	BudgetFlow	16/30 (53.3%)	\$9.95	18.00	1.81
	learned task-router	15/30 (50.0%)	\$9.95	17.00	1.71
	strong-model-only	15/30 (50.0%)	\$9.95	16.50	1.66

Table 4. Main paid evidence. Each row reports the leader for one 30-task batch. Resolved Rate, TRV, and TRV per Dollar are the three primary metrics. Red bold entries mark BudgetFlow frontier signals; blue bold entries mark the strongest control in boundary runs. The 4-policy BudgetFlow row uses total observed spend, including non-scoreable abort overhead.

Mode	Operating condition	Evidence	Interpretation
Value Divergence	Task values differ and model gains are selective.	In the 4-policy case, BudgetFlow resolves 14/30 but reaches TRV 18.50 and 2.32 TRV/\$, above strong-model-only’s TRV 18.00 and 1.90 TRV/\$.	Value-aware allocation moves budget toward higher-value resolved tasks.
Strong Coverage	The strong model covers the batch broadly under the cap.	In the strong boundary case, strong-model-only reaches 17/30, TRV 20.00, and 2.01 TRV/\$, above BudgetFlow’s 15/30, TRV 17.50, and 1.76 TRV/\$.	The strong model itself forms a frontier boundary.
Model Fit	The learned routing signal places model tiers accurately.	In the router diagnostic, learned routing reaches 19/30, TRV 21.00, and 2.11 TRV/\$, with stronger T3-start placement than BudgetFlow.	Future BudgetFlow policies should absorb the model-fit signal while retaining Task Value and Shared Budget governance.

Table 5. Frontier mode analysis. Each mode explains which signal limits or moves the observed cost-value frontier under the same batch-level objective.

Policy	Resolved Rate	TRV	TRV/\$
BudgetFlow	16/30 (53.3%)	18.00	1.81
learned task-router	15/30 (50.0%)	17.00	1.71
strong-model-only	15/30 (50.0%)	16.50	1.66
budget-only	12/30 (40.0%)	13.00	1.31
cheap-model-only	11/30 (36.7%)	12.00	1.21

Table 6. Latest audited 5-policy run over 30 tasks.

Value profile	BF TRV	Control TRV	Margin
equal	16.00	15.00	+1.00
criticality	18.00	17.00	+1.00
compressed	17.00	16.00	+1.00
expanded	20.00	19.00	+1.00

Table 7. Task Value sensitivity on the latest audited run.

7.2. Budget Sensitivity

Budget sensitivity is the closest analogue to a frontier plot. Figure 2 fixes observed outcomes and replays budget accounting at different caps. It therefore describes the observed cost-value frontier for this completed run. The strong-model-only policy is strongest at some low and mid caps, while BudgetFlow reaches the observed frontier at the full cap. The important reading is the frontier shape: the useful policy changes with budget pressure, and BudgetFlow provides the interface that makes this movement reportable.

7.3. KV Cache Cost-Discount Sensitivity

KV Cache Cost-Discount sensitivity tests a serving-cost question. Modern serving substrates can make repeated input context cheaper; this changes effective spend while leaving verified outcomes unchanged. Table 8 re-costs repeated input tokens under KV-cache discounts. BudgetFlow leads or stays close through KV90. At KV98 and KV99, the learned task-router has the highest TRV per Dollar, while BudgetFlow keeps a higher TRV than the learned task-router in the fixed-outcome table.

This pattern gives a useful serving-cost reading. BudgetFlow gains part of its efficiency by allocating scarce budget across tasks. When cache discounts become extreme, the

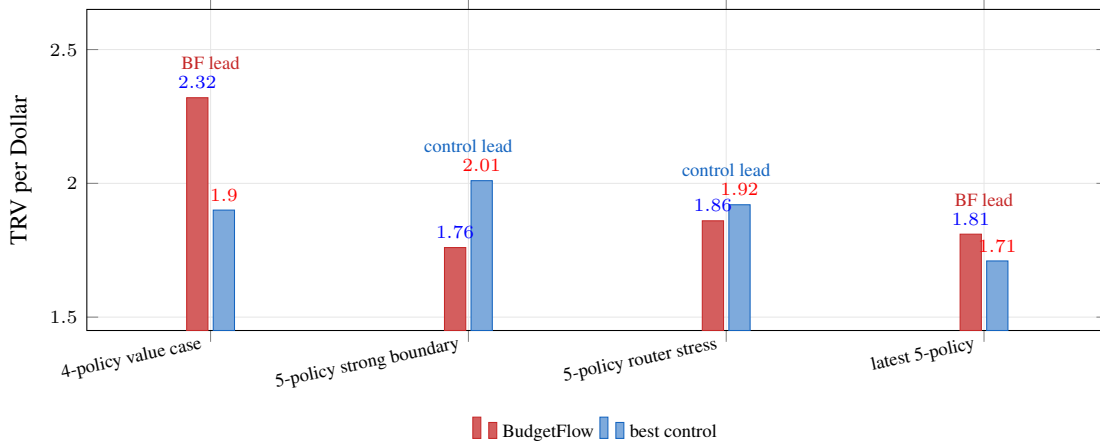


Figure 1. Multi-run main evidence. Each pair compares BudgetFlow with the best available control in that run on TRV per Dollar. The figure separates the repeated-run evidence from the sensitivity analysis in Section 7.

KV profile	BF	Best control	BF margin
KV0	1.81	1.71	+0.10
KV50	3.26	3.14	+0.11
KV90	9.06	8.98	+0.09
KV98	14.08	14.28	-0.20
KV99	15.13	15.41	-0.29

Table 8. KV Cache Cost-Discount sensitivity. Values are TRV per Dollar.

cost penalty of repeated or stronger model use shrinks, and value-blind controls can become more competitive on TRV per Dollar. The result makes the serving boundary explicit: BudgetFlow is most useful under meaningful budget pressure, and serving optimizations can shift which policy lies on the efficiency frontier.

8. Discussion and Future Work

Formulation-level generalization. The SWE-bench experiments instantiate the formulation in a verified coding domain. The formulation applies to task domains where tasks expose Task Value, Estimated Token Cost, Model Fit, and a Verifier. Coding tasks use tests as the Verifier. Support tasks can use policy checks and service outcomes. Spreadsheet tasks can use cell-level checks or rule engines. Research and document-production tasks can use human review or domain checks. The calibration of each interface is domain-specific; the allocation objective and budget ledger are shared.

Frontier-aware reporting. The same BudgetFlow framework explains multiple frontier positions across the paid runs. Value Divergence favors value-aware allocation, Strong Coverage lets the strong-model-only baseline define a boundary, and Model Fit lets a learned task-router

approach or form a boundary. Reporting the frontier turns these cases into an operating guide for shared-budget agent batches.

Future work. Fine-grained task execution control. BudgetFlow can move from task-level allocation to finer-grained execution control inside a task. The current paper assigns model opportunities at the task level. A richer policy could decide which task stage or segment deserves the strong model, how much repair runway to preserve, and when stop/defer decisions should be made inside a long attempt. This direction should be evaluated against task-level controls because finer granularity can improve allocation only when the stage signal is reliable enough to justify the added switching cost.

Continual policy learning with auditability. Completed batches produce cost traces, routing outcomes, verifier results, and frontier positions. A future policy can use those records to update future caps, route choices, and stop/continue decisions. The important systems constraint is that each learned update should remain inspectable: the next allocation should be explainable through the same value ledger, Estimated Token Cost estimate, model-fit evidence, Shared Budget state, and verifier-settled outcome history.

Serving-aware and multi-tenant BudgetFlow. Serving systems such as vLLM, SGLang, and Dynamo optimize execution substrates: batching, cache locality, scheduling, and inference throughput. BudgetFlow operates at the task-budget layer above that substrate. A serving-aware BudgetFlow could pass task value, Shared Budget state, Estimated Token Cost, and outcome history into scheduling hints such as priority, continuation, model opportunity, and cache-sticky execution. The same interface can then extend from one budget owner to multiple teams, users, or services that share an agent execution substrate with distinct budgets

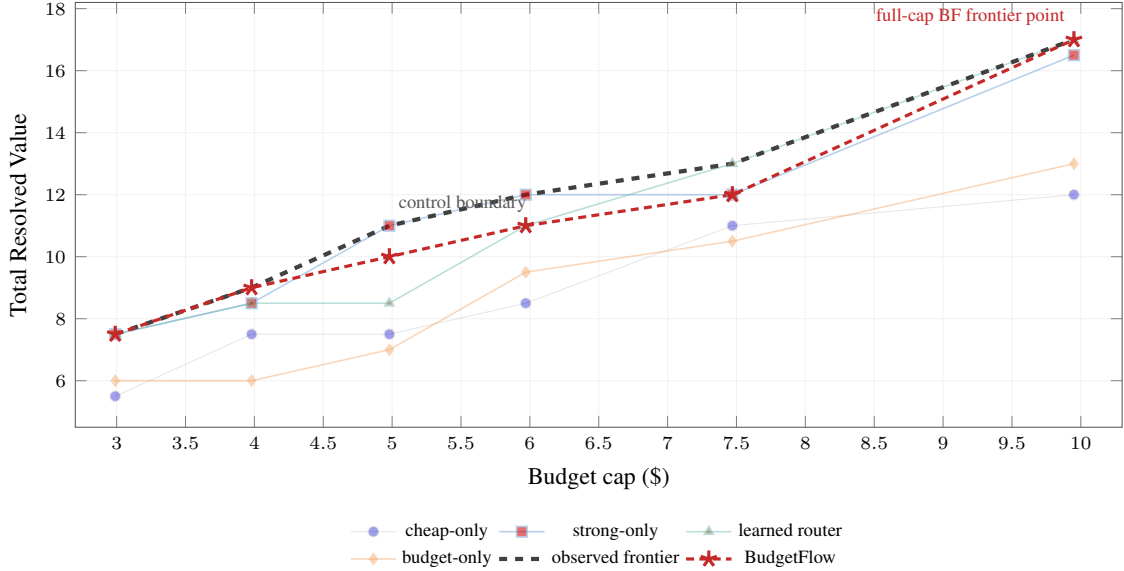


Figure 2. Budget sensitivity diagnostic on the latest audited run. Outcomes are fixed; budget accounting is replayed at alternate caps. The dashed line is the observed frontier; BudgetFlow is emphasized in red, while controls are shown as lighter diagnostic curves.

and priorities. **This turns Batch-Level Task Budget Allocation into a candidate control plane for multi-tenant agent workloads.**

9. Conclusion

This paper formulates Batch-Level Task Budget Allocation for multi-step agent tasks under a shared hard budget. The core question is which task should receive the next scarce model opportunity when tasks have different value, different estimated token cost, and different model-tier fit. BudgetFlow answers this question with a simple auditable interface: Task Value, Estimated Token Cost, Model Fit, Shared Budget state, and a Verifier-settled outcome.

The practical role of TRV is value accounting. It is a frozen value score that asks whether the same spend resolves more pre-declared task value. This makes mixed outcomes meaningful: the paper reports which policy lies on the cost-value frontier under a Shared Budget, instead of reducing the evidence to one point comparison.

The SWE-bench experiments instantiate this interface in a verified coding domain. In the latest audited 5-policy run, BudgetFlow **improves Resolved Rate, TRV, and TRV per Dollar**. Across the broader paid evidence, BudgetFlow also explains frontier modes induced by Value Divergence, Strong Coverage, and Model Fit. It is most useful when task values are uneven, model-tier advantage is task-specific, and the Shared Budget forces tradeoffs across tasks. Strong-model-only defines a boundary when the strong model is broadly cost-effective under the cap. A learned task-router defines a strong operating point when Model Fit explains

most of the useful routing signal. The sensitivity analyses show how the frontier moves under alternate value profiles, budget caps, and serving cost assumptions.

The broader value of BudgetFlow is the formulation. Coding tasks provide a strict verifier and measurable spend, but the same batch-budget structure appears whenever many agent tasks compete for one resource pool and produce outcomes with unequal value. By **separating allocation signals from outcome settlement**, BudgetFlow offers a portable way to reason about verified value under fixed resource constraints, from SWE-bench-style coding to future serving-aware and multi-tenant agent systems.

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